



Armada Kalamunda Group

Premature Pre-labour Rupture of Membranes Guideline

1. Purpose

- To reduce the risks associated with Premature Pre-labour Rupture of Membranes (PPROM) that relate to maternal and neonatal infection, cord prolapse, and fetal compromise and co-morbidities associated with PPRM and premature birth.
- To stabilise threatened preterm labour when associated with PPRM, administer the first dose of corticosteroid and then transfer in-utero to a tertiary centre with suitable neonatal intensive care facilities (NICU) according to gestation.
- Counsel the woman about active management and commencement of antibiotic therapy on confirmation of PPRM with a positive Group B Streptococcus (GBS) status between 35 -36 +6 weeks gestation. If woman is 37 weeks exactly, refer to the PROM≥37weeks guideline.
- Counsel the woman about increased risk of chorioamnionitis between 34 – 36 weeks and decide on active or conservative management with the woman.

2. Applicability

This guideline is applicable for midwifery, medical and nursing staff providing care for patients presenting to Armada Health Service with premature pre-labour rupture of membranes.

3. Background

PPROM occurs in 2% of pregnancies but is associated with 40% of preterm births and can result in significant neonatal morbidity and mortality.

Three main causes of neonatal death associated with PPRM are:

- Prematurity
- Sepsis
- Pulmonary hypoplasia

Outcome for preterm neonates depend on the place of birth and access to neonatal intensive care facilities.

3.1 Associated risks of PPRM

- Preterm labour
- Cord prolapses
- Placental abruption
- Intrauterine infection / amnionitis
- Pulmonary hypoplasia

- Limb positioning defects
- Perinatal mortality

4. Clinical guidance for PPRM

Premature Prelabour Rupture of Membranes PPRM <37 weeks gestation. Refer to Appendix: PPRM Flowchart.

4.1 Initial Telephone Consultation

- When a woman telephones with a history of suspected PPRM, confirm her gestation and ask her to use a sanitary pad before presenting to Antenatal Assessment Unit (AAU).

4.2 Assessment, examination, investigations and diagnosis

4.2.1. Key components of initial assessment

- Confirmation of Rupture of Membranes (ROM) including assessment for differential diagnoses
- Confirmation of gestation
- Assessment of maternal wellbeing
- Assessment of fetal viability and well being
- Document time and history of the reported vaginal loss. Note type, colour, amount, and any abnormal smelling discharge. Obtain swabs for Group B Streptococcus (GBS).
- Document maternal temperature, pulse and blood pressure (BP), respiratory rate and oxygen saturation and calculate MADDS score.
- Perform an abdominal examination noting:
 - Fundal Height (FH)
 - Lie (if appropriate depending on gestation)
 - Presentation
 - Uterine tenderness, irritability/activity
- Auscultate the fetal heart and confirm presence of fetal movements

4.2.2. Speculum examination with consent.

- Offer the woman to empty her bladder prior to sterile speculum examination.
 - Exclude cord prolapse
 - Visualise pooling of liquor (note any presence of vernix)
 - Collect low and high vaginal swabs for microscopy and culture
 - Use Amnisure to help confirm the diagnosis if no obvious liquor seen
- If women gives history of leaking for more than 12 hours, the test might be false negative and gives false reassurance

4.3 Fetal assessment

- Cardiotocography (CTG), depending on period of gestation (POG), to assess fetal wellbeing
- Bedside ultrasound to assess liquor volume (and visualise presentation) if clinical picture allows

- Consider formal ultrasound for fetal number, presentation and Maximum Vertical Pocket of liquor (MVP)/Amniotic fluid index (AFI)

4.4 Other maternal investigations

- C-Reactive Protein (CRP)– twice weekly
- Full blood picture – twice weekly
- Midstream specimen of urine for bacteriology at initial presentation and later if symptoms suggestive of evolving urinary tract infection
- High Vaginal Swab (HVS), Low Vaginal Swab(LVS)

When escalating care e.g., Reviewing bloods, USS or CTG concerns. The reporting midwife MUST co-sign with the medical officer the acknowledgement of results reviewed and/or any follow up care planned.

4.5 Contractions in gestation (23 – 34+6)

- If contracting, aim to stabilise and administer first dose of corticosteroid and then transfer in-utero to a tertiary centre.
- Between 23-29+6 transfer to KEMH and between 30 – 34 weeks to transfer to FSH

If Contracting

- (Refer to Threatened Preterm labour Flowchart)

Administration of Corticosteroids if 24 to 34 weeks

Aim to administer first dose before transfer to tertiary hospital.

- For prophylaxis against neonatal respiratory distress syndrome, to accelerate lung maturation and to decrease the risk of neonatal intracerebral haemorrhage and necrotising enterocolitis, a course of Betamethasone (Celestone) is given to the women with a gestation of between 24 to 34 weeks and 6 days.
- Two intramuscular injections of 11.4mg of Betamethasone are to be given 24 hours apart.
- No repeat courses are given. In the event of maternal diabetes, the administration of corticosteroids may affect blood glucose level control.

4.6 In-Utero Fetal Transfer

- There is clear evidence that neonatal outcome is best if a significantly preterm infant is born in the hospital which provides the appropriate level of neonatal care, when compared with infants requiring transfer after birth.
- In-utero transfer should not, however, be undertaken if there is significant risk of birth occurring during transfer.

4.7 Expectant management below 35 weeks gestation (23 –34+6)

- When PPROM confirmed, the woman is admitted and observed for 24-48hrs in case of threatened preterm labour.
- Commence oral Erythromycin 250mg 4 times a day for 10 days or until delivery if this occurs earlier in the absence of any signs of infection.

- May go home after 24-48hrs on oral erythromycin with an observation chart to record her temperature every 4 hours and to observe vaginal loss, fetal movements and uterine contractions.
- Give the woman disposable thermometers and self-observation chart.
- Women are reviewed weekly in AAU.
- Await spontaneous onset of labour until 37 completed weeks of gestation when the woman will be counselled by a Specialist Obstetrician about any risks associated with continuing the pregnancy
- Advise the woman to return to the hospital if she develops fever ($>37.0^{\circ}\text{C}$), the colour or the odour of vaginal loss changes, the baby does not move as much as previously, or contractions start.

4.8 Antibiotic Prophylaxis

- Studies show that prophylactic antibiotics prolong and reduce maternal and neonatal sepsis.
- Commence oral Erythromycin 250 mg 4 times a day for 10 days or until delivery if this occurs earlier in the absence of any signs of infection.

4.9 Intrapartum

- Benzylpenicillin 3 grams IV stat, followed by 1.8 grams Benzylpenicillin every 4 hours until birth
- If allergic to Penicillin, give Clindamycin 900mg stat dose IV followed by 900mg IV every 8 hours

4.10 Signs and Symptoms of Chorioamnionitis

Diagnosis relies on clinical presentation and / or laboratory investigations. Specialist Obstetrician to review to determine management in following situation: Do not inhibit labour if there are signs of chorioamnionitis.

- Maternal fever $> 38^{\circ}\text{C}$ on two separate occasions with any 2 of the following
- Maternal tachycardia ($> 100\text{ bpm}$)
- Fetal tachycardia ($>160\text{ bpm}$)
- Uterine tenderness
- Offensive smelling vaginal discharge
- Increased white cell count ($>15,000$)
- C-Reactive Protein >40

4.11 PPROM at $\geq 35\text{-}36\text{+}6$ weeks gestation

All investigations as above.

- Studies are currently in progress to establish whether to recommend expectant or active management for women with PPROM between 34 to 36 completed weeks of gestation.

4.12 Active or Conservative Management

- The woman and her partner need to be counselled about the management options of active or conservative management for PPRM as detailed below and in relation to the woman's clinical condition, fetal wellbeing and GBS status.

4.13 Active management (allowing / encourage birth to proceed)

- If in established labour
- There are signs and symptoms of chorioamnionitis present
- Significant antepartum haemorrhage
- If GBS positive
- Signs of fetal compromise
- Consider caesarean section if birth is not imminent and fetus showing signs of distress
- If the woman chooses active management after being fully informed

4.14 Induction/augmentation

- Prostaglandin Induction if Bishop Score below 6 or cervix deemed unfavourable
- If cervix favourable or Bishop Score greater than 6, commence syntocinon infusion.

4.15 Expectant management

- When PPRM confirmed, the woman is admitted and observed for 24-48 hrs in case of threatened preterm labour.
- Commence oral Erythromycin 250 mg 4 times a day for 10 days or until delivery if this occurs earlier in the absence of any signs of infection.
- May go home after 24-48hrs on oral Erythromycin with an observation chart to record her temperature every 4 hours (Appendix 2) and to observe vaginal loss, fetal movements and uterine contractions.
- Woman should be advised to avoid vaginal intercourse, the use of tampons and swimming/bathing.
- Give the woman disposable thermometers and self-observation chart.
- Await spontaneous onset of labour until 37 completed weeks of gestation when the woman will be counselled by a Specialist Obstetrician about any risks associated with continuing the pregnancy.
- Advise the woman to return to the hospital if she develops a fever ($> 37.0^{\circ}\text{C}$), the colour or the odour of vaginal loss changes, the baby does not move as much as previously, or contractions start.

4.16 Antibiotic Prophylaxis Management

- Studies show that prophylactic antibiotics prolong pregnancy and reduce maternal and neonatal sepsis
- Commence oral Erythromycin 250 mg 4 times a day for 10 days or until delivery if this occurs earlier in the absence of any signs of infection.
- Intrapartum,
- Benzylpenicillin 3 Grams IV stat followed by 1.8 grams Benzylpenicillin every 4 hours until birth

- If allergic to Penicillin, give Clindamycin 900 mg IV every 8 hours

4.17 Progesterone

Progesterone should not be commenced in women with PPROM and should be discontinued in women using it prior to PPROM.

4.18 Signs and Symptoms of Chorioamnionitis

Diagnosis relies on clinical presentation and / or laboratory investigations.

- Maternal fever > 38° C on two separate occasions* with any 2 of the following.
- Maternal tachycardia (> 100 bpm)
- Fetal tachycardia (> 160 bpm)
- Uterine tenderness
- Offensive smelling vaginal discharge
- Increased white cell count (>15,000)
- C-Reactive Protein >40

4.19 Treatment of suspected/confirmed Chorioamnionitis

- Commence Amoxycillin 2g IV initial dose then 1g IV every 4 hours,
- Gentamicin 5 mg / kg IV daily and
- Metronidazole 500 mg IV every 12 hours.
- If patient not fasting or able to take oral diet, aim for oral Metronidazole 400mg twice in a day.
- **Do not** inhibit labour.
- Consider optimal mode of deliver under antibiotic therapy. (LSCS versus vaginal birth) on the basis of the clinical picture and the anticipated duration until birth

4.20 Placenta and membranes

- Histological examination of placenta and membranes with evidence of acute inflammation may confirm diagnosis post birth.
- Placental swabs to be taken for culture sensitivity and gram stain

4.21 Postnatal maternal antibiotics

- May change to oral antibiotics once the woman is afebrile and tolerating oral medication e.g. Amoxycillin 500 mg every 8 hours and metronidazole 400 mg every 12 hours or amoxicillin / clavulanic acid (Augmentin Duo Forte x 1 every 12 hours) for the rest of the 5 days.

4.22 Neonatal care

- Refer to Paediatric presence at birth.
- Routine observations for possible sepsis of the neonate as for GBS positive woman.

5. Legislative context

The following documents are required to give effect to this policy:

- East Metropolitan Health Service (EMHS) [Fetal Surveillance and Cardiotocography \(CTG\) Policy EMHS: 195](#)
- EMHS [Consent Policy EMHS: 5](#)

6. Supporting information

The following documents support the implementation of this policy:

- AKG [Group B Streptococcus Guideline](#)
- AKG [Antimicrobial Stewardship Policy](#)
- AKG [Medication Management Procedure](#)

Compliance, monitoring and evaluation

Compliance with this policy will be monitored via:

- Monitoring of reported clinical incident data via Datix CIMS.

Department specific findings are to be reported at the departmental meeting for review and identification of any opportunities for improvement.

7. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 3 – Action 3.15 Antimicrobial stewardship
- Standard 5 – Action 5.11 Clinical Assessment
- Standard 6 – Action 6.9 Communication critical information
- Standard 6 – Action 6.7 Clinical Handover

8. References

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4. Morris JM, Roberts CL, Bowen JPJ, Bond DM, Algert CS, Thornton JG, Crowther CA. Immediate delivery compared with expectant management after preterm pre-labour rupture of the membranes close to term (PPROMT trial): a randomised controlled trial. Lancet 2016;387:444–52
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7. Bond DM, Middleton P, Levett KM, van der Ham DP, Crowther CA, Buchanan SL, et al. Planned early birth versus expectant management for women with preterm prelabour rupture of membranes prior to 37 weeks' gestation for improving pregnancy outcome. Cochrane Database of Syst Rev 2017;:

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9. King Edward Memorial Hospital. Rupture of Membranes – Spontaneous: Preterm Prelabour Rupture of Membranes (PPROM) 23-37 weeks. Obstetrics & Gynaecology Clinical Practice Guideline; November 2019
10. Bond DM, Middleton P, Levett KM, van der Ham DP, Crowther CA, Buchanan SL, et al. Planned early birth versus expectant management for women with preterm prelabour rupture of membranes prior to 37 weeks' gestation for improving pregnancy outcome. Cochrane Database of Systematic Reviews. 2017 (3). Available from: <http://dx.doi.org/10.1002/14651858.CD004735.pub4>
11. Singh K, Mercer B. Antibiotics after preterm premature rupture of the membranes. Clinical Obstetrics and Gynecology. 2011;54(2):344-50.
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9. Glossary

The following definitions are relevant to this policy.

Term	Definition
AAU	Antenatal Assessment Unit – Outpatient clinic
GBS	Group B Streptococcus (GBS) status , may be referred to as “GBS positive”
IOL	Induction of Labour (IOL)
PPROM	Preterm Pre-labour Rupture of Membranes (PPROM)
PROM	Premature Rupture of Membranes (PROM)
RUPTURE	Rupture of the fetal membranes before 37+0 completed weeks of pregnancy (i.e. preterm) and before the onset of labour (i.e. prelabour)

10. Document contact

Enquiries relating to this guideline may be directed to:

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11. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

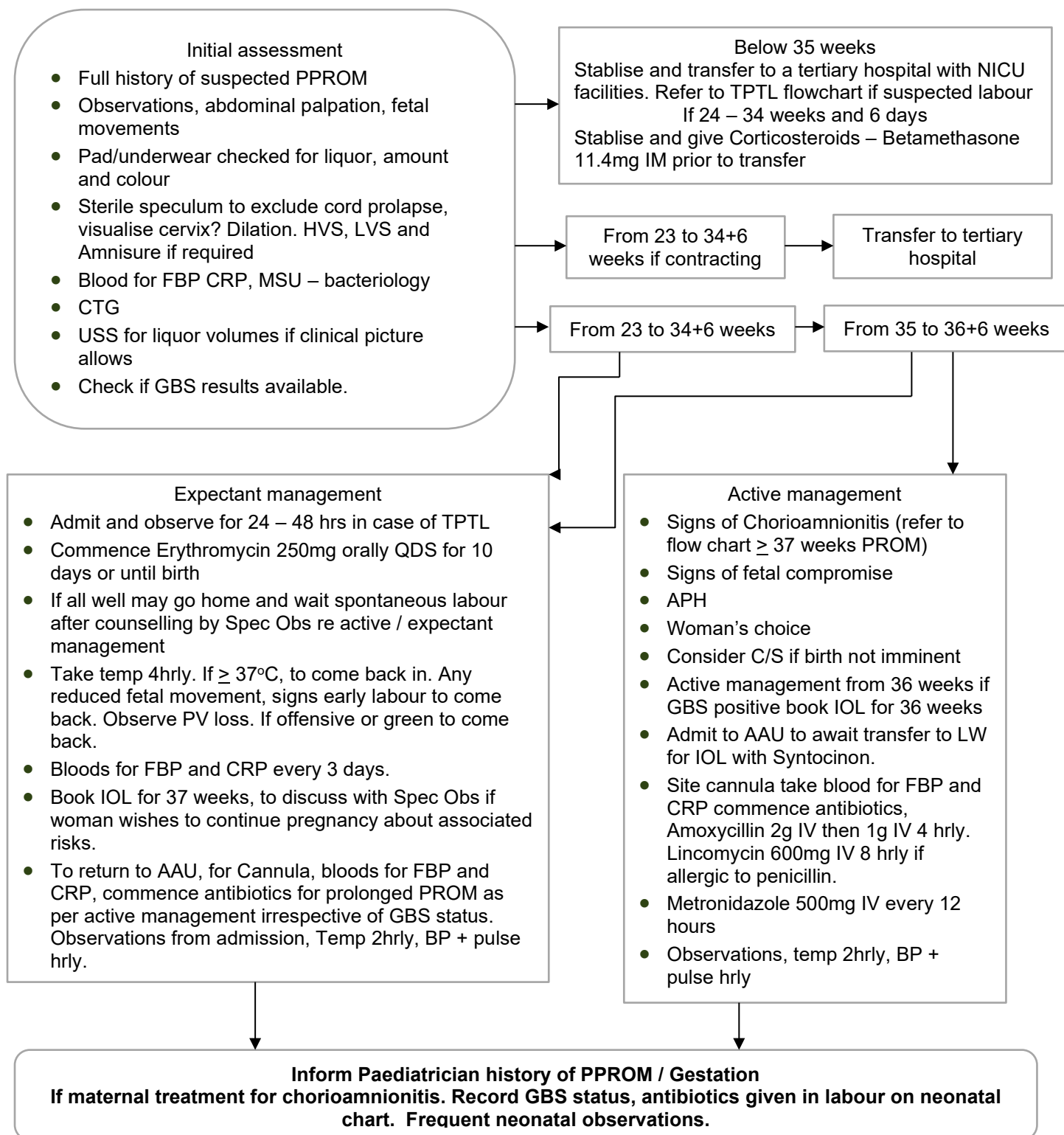
Version	Effective from	Effective to	Amendment(s)
V5	31/01/2024	31/01/2027	Scheduled review

12. Authorisation

This guideline has been authorised and issued by the Director Clinical Services.

Authorised by	Dr Tania Strickland – A/Director Clinical Services
Authorisation date	30/01/2024
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Appendix 1: PPROM $\leq$ 36.6 weeks flowchart



Appendix 2: Home observation chart for women with conservation management for PPRM

- If temperature over 37.5°C on two occasions, come into Antenatal Assessment Unit (AAU)
- If temperature is 38°C come straight into AAU (Tel :08 9391 2947)
- If vaginal loss becomes red, green or smelly come to AAU (Tel: 08 9391 2947)
- If the movements of your baby are slowing down, come to AAU (Tel: 08 9391 2947)

Date	Time	Temperature	Vaginal Loss	Fetal movements
	6 am			
	10 am			
	2 pm			
	6 pm			
	10 pm			
	6 am			
	10 am			
	2 pm			
	6 pm			
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